

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE

Final Exam, Spring 2022

Course Code: PHY 1201

Course Title: Physics-II

Time: 2 hours 30 minutes

Full Marks: 50

(i) The figures in the right-hand margin indicate full marks

(ii) Course Outcomes and Bloom's Level are mentioned in additional Columns

Part A

[Answer the questions from the followings]

1. a) Distinction between metal, insulator and semiconductor in terms of energy band diagram. CO1 U 5
- Or,
1. a) Define: i) Lattice ii) Basis iii) Crystal iv) Unit Cell v) Crystalline Solid CO1 R 5
1. b) Show that the packing fraction for a fcc lattice is ~ 0.74 and hence write the meaning of this number. CO2 A 5
2. a) Define Miller indices and draw the plane of (100), (010), (101) and (111). CO1 U 5
2. b) Show that in a crystal of cubic structure, the distance between the planes with Miller indices h, k, l is equal to $d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$, where a is the lattice parameter. CO2 A 5
- Or,
2. b) If a plane of a cubic lattice intersects the X, Y and Z axes at 3, 2 and 2 respectively. If lattice parameter (X+1) and find the distance of the plane from origin and draw this plane. Here X is the last digit of your student ID. CO2 E 5

Part B
[Answer the questions from the followings]

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|-----|----|---|-----|---|---|
| 3. | a) | Explain "A moving clock always appears to go slow". | CO2 | A | 6 |
| 3. | b) | The length of a spaceship is measured to be exactly half its actual length. Calculate
(i) the speed of the spaceship
(ii) the time dilation corresponding to one second on the spaceship. | CO2 | A | 4 |
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| 4. | a) | State the laws of radioactive disintegration. Show that, the number of radioactive atoms decreases exponentially with time. | CO2 | A | 5 |
| 4. | b) | The half-life of a radioactive substance is 30 days. Calculate (i) the radioactive decay constant, (ii) the mean life and (iii) the time taken for $(1/3)$ of the original number of atoms to remain unchanged. | CO2 | E | 5 |
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| 5. | a) | Define 'Nuclear Fission' and 'Nuclear Fusion' with an example of each. | CO1 | A | 4 |
| 5. | b) | Find the nuclear Mass, Size, density, binding energy and binding energy per nucleon for ${}_{88}\text{Ra}^{226}$. | CO2 | A | 6 |
| Or, | | | | | |
| 5. | a) | Write short note on:
(i) Photoelectric effect
(ii) Compton effect | CO1 | R | 4 |
| 5. | b) | Show that, the mean life of a radioactive atom is the reciprocal of the radioactive constant. | CO2 | R | 6 |